THEME 3 · RISING PLASTIC POLLUTION IN COASTAL CITIES

StarDust

Is the trash barrier still doing its job — or does a crew need to go now?

A low-cost monitoring retrofit for existing drainage trash barriers. It reads the water-level difference across the barrier and the tension in one mooring line, checks whether the data is valid, and tells the maintenance team when an inspection is worth a visit.

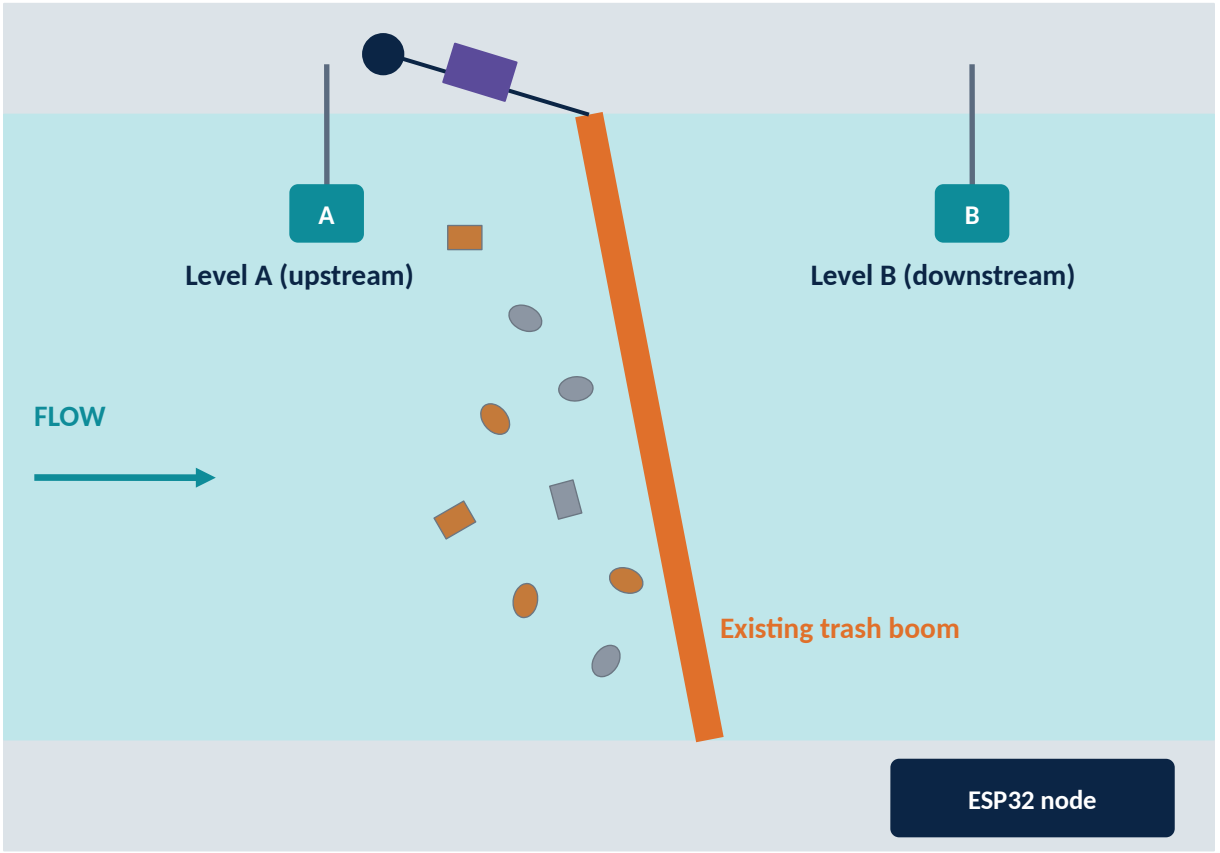
Team: [Team name] **Team ID:** [ID]

Members: [Member 1], [Member 2], [Member 3], [Member 4]

Institute: [Institution]

Contact: [email] · [phone]

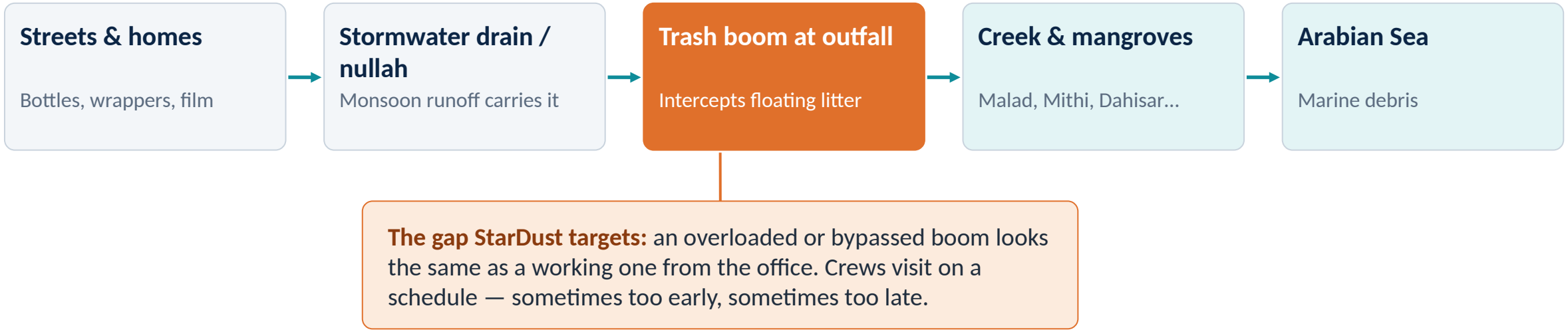
Proposed arrangement on an existing barrier (plan view)



Conceptual illustration only. Not an installed system, not a surveyed site, not to scale.

THE PROBLEM

Booms catch drain plastic, but no one sees them between visits



≈ 5,000 t / yr

plastic estimated to flow from Mumbai's waterways and nullahs to the Arabian Sea

InnovateX Theme 3 brief (survey estimate quoted by organisers)

₹14.5 crore

2-year BMC contract (Jul 2026) to operate & maintain trash booms on six rivers/nullahs in the western suburbs

Free Press Journal, 30 Jul 2026; booms commissioned Jul 2022 after NGT directions

Visits misdirected

Industry note: scheduled screen cleaning often visits clear screens and misses blocked ones; uncleared screens add flood risk

Aquasition, trash-screen monitoring (2019)

Users: drainage engineer (reviews evidence) · clearance crew (inspects, clears) · asset owner / O&M contractor (pays for visits). Candidate area: Malad Creek catchment — outlet, barrier presence and permissions NOT yet surveyed.

EXISTING SOLUTIONS

The sensors exist. Does combining them make better decisions?

Method	What it observes	Already used?	Where it falls short for drain booms
Scheduled manual inspection	Direct visual condition	Yes — current practice	Only at visit time; visits can be mistimed
Single upstream level	Water level rising	Yes — commercial screens	Rain or tide raise level with no debris
Differential level (Δh)	Head across screen	Yes — Aquasition, AMETEK	Built for fixed screens; tide/backwater confound; shallow booms give small Δh
Camera monitoring	Images of the screen	Yes — commercial	Needs a person to view; night, glare, no load info
Mooring-line tension	Mechanical load on boom	Research — tank & field tests	Flow speed alone changes load; mostly design studies, not alerts
StarDust (proposed)	Δh + tension + validity	To be tested	Must beat level-only & tension-only to justify itself

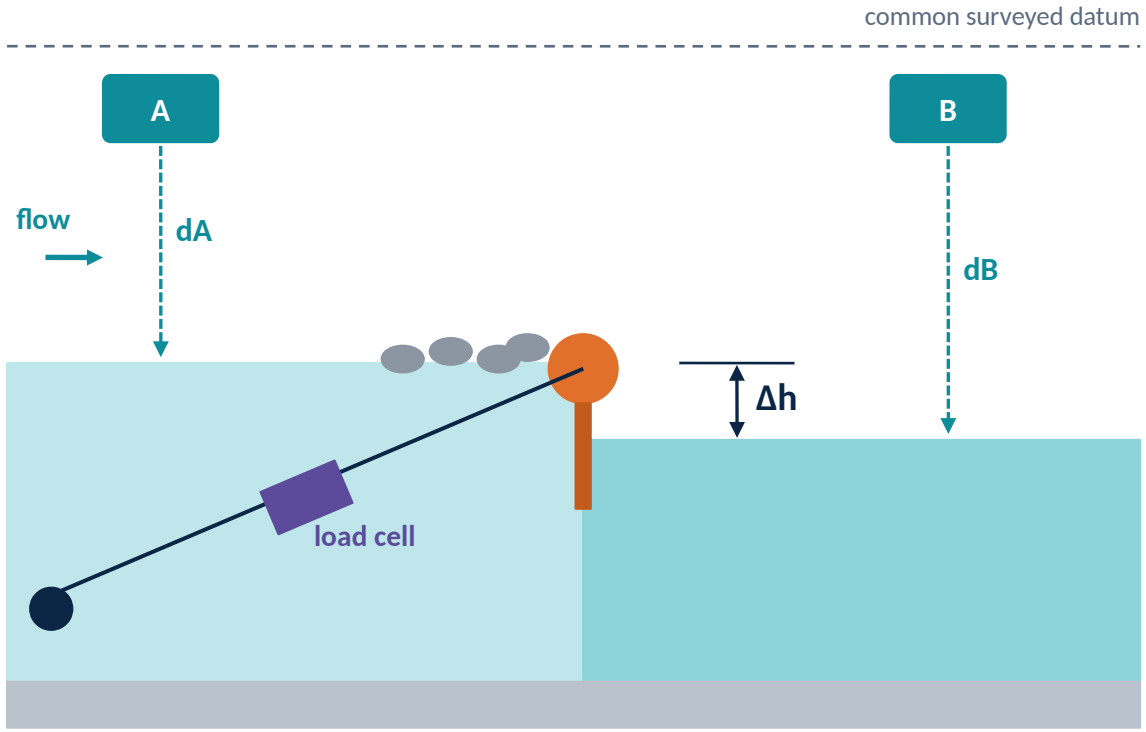
Not our claim
ESP32, ultrasonic level, load cells or 'sensor fusion' being new. We found related products and research.

Our proposed contribution
Site-calibrated decision logic that keeps fault and uncertainty states, and a fair test against level-only and tension-only monitoring.

Aquasition (2019) trash-screen monitoring; AMETEK app note, differential measurement for traveling screens; IPT towing-tank trash-boom tests (IAHR); trash-net mooring studies (Ocean Engineering 2022, Fluids 2022). Full list in appendix.

Two hydraulic + one mechanical reading → one evidence-based state

Side view across the barrier



h = z − d Δh = hA − hB

T [N] = k × (raw − zero offset) — calibrated on the actual fixture

Data path

Level A — upstream
ultrasonic, JSN-SR04T-class

Level B — downstream
ultrasonic, sequential trigger

Mooring load cell → HX711
24-bit bridge ADC, DOUT/SCK

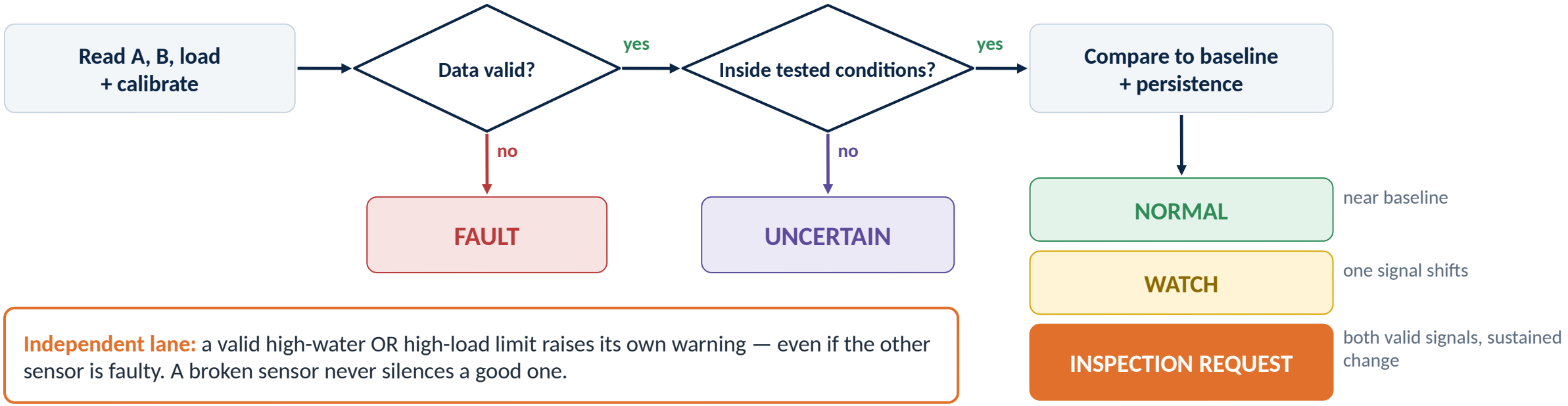
ESP32
- timestamps & calibration
- range / stale / jump checks
- compare with clean baseline
- persistence rules → state

State + raw log + alert
microSD / laptop · LED/buzzer · Wi-Fi optional

Human operator
reviews · crew inspects · verifies recovery

Bench design. Field version needs rated load path, enclosure, power and permission.

Evidence, not diagnosis: StarDust is allowed to say “uncertain”



Independent lane: a valid high-water OR high-load limit raises its own warning — even if the other sensor is faulty. A broken sensor never silences a good one.

Why one signal can mislead (illustrative, not measured)

Flow surge, clean boom: tension rises, Δh barely moves → WATCH, not inspection.

High tide / backwater: both levels rise, Δh can shrink, tension steady → UNCERTAIN.

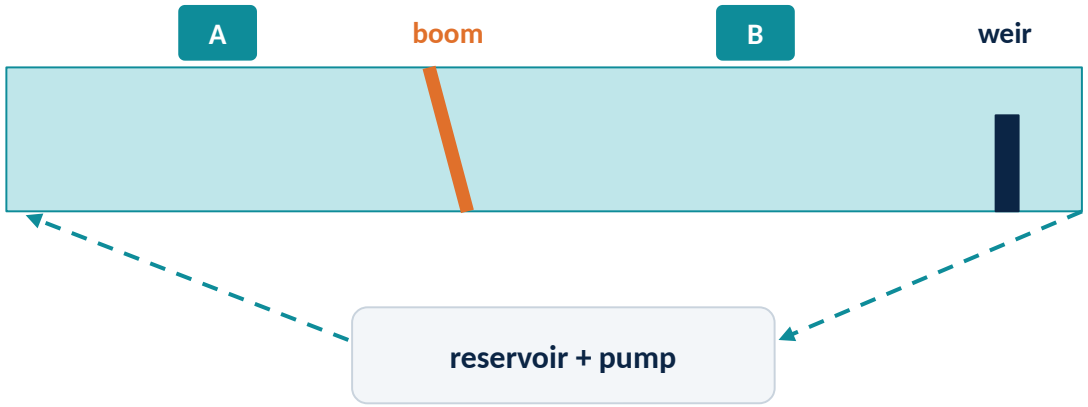
Debris build-up: Δh and tension both stay high for the persistence window → INSPECTION REQUEST.

System boundary — StarDust does NOT

- remove waste or open the boom
- identify plastic or weigh it
- guarantee flood prevention
- use un-validated thresholds or AI

A ₹9,950 bench rig that can prove — or disprove — the idea

Tabletop recirculating channel (planned)



A	B	C	D	E
Clean flow	**Add debris**	**Change flow**	**Backwater**	**Fault + recovery**
baseline	same flow	clean boom	raise weir	unplug, clear

Fair comparison: freeze rules first, then run level-only, tension-only and combined StarDust on the same held-out runs. Ground truth set independently (photos, rulers, known debris), never by StarDust itself. 60-run matrix in appendix.

What we will report

- False-positive rate
- Missed-event rate
- Detection delay
- Uncertain-output rate
- Valid-data fraction

Bench item (planning allowance)	INR
ESP32, 2 level modules, load cell, HX711, logging, wiring	3,400
Transparent channel + leak containment	1,800
Pump, hoses, reservoir	1,700
Boom, mooring, frame · weir · reference tools · misc	1,750
Base total	**8,650**
15% reserve → planned total (rounded)	**9,950**

Go / no-go gate: if Δh and tension changes are not clearly bigger than sensor noise, or combined logic does not beat the simpler methods, we simplify or stop.

Budget = engineering planning estimates, not supplier quotes; borrowed laptop and tools assumed; excludes any field installation. All tests are PLANNED — no results yet.

Bench evidence → one measured pilot → scale only what works

Bench verification
rig + 3-method test

Permitted site survey
Malad catchment candidate

One-site pilot
with named operator

Measured evaluation
vs baseline period

Potential expansion
per-site calibration

Each arrow is a gate: we move right only on evidence. Nothing beyond stage 1 is arranged yet.

Operator workflow

1 Condition flag
↓
2 Engineer review
↓
3 Crew inspection
↓
4 Clear if needed
↓
5 Recovery verified

What a pilot will actually measure

- Alert-to-inspection time (min)
- Inspections & unnecessary visits
- False alerts & missed events
- Plastic weighed independently (kg)
- Disposal destination
- Operating & servicing cost

Unresolved conditions stay open. An acknowledgement is not proof of clearance.

Prospective value (to be measured)

- Better-timed visits across many booms
- Booms kept intercepting during storms
- An evidence trail for O&M contracts
- Scales by per-site calibration, not one threshold

We are not promising a cleaner sea. We are offering a cheap, testable way to know when a boom needs a crew — and the evidence to decide if it is worth it.

Sensor and controller interfaces (bench)

Block	Interface	Design note (to verify)
Level A / B: JSN-SR04T-class ultrasonic	TRIG / ECHO GPIO, triggered one after the other	5 V echo → divider or level shifter for 3.3 V ESP32; check blind zone & beam footprint
Load cell + HX711	DOUT / SCK (2-wire serial)	Fixture sets loading direction; capacity & overload stop; zero/span, creep checks
ESP32 dev board	3.3 V logic, Wi-Fi optional	Timestamps, sequence numbers, validity flags, state machine
Logging	microSD over SPI, or USB serial to laptop	Raw + calibrated values, rule & calibration version
Local indication	GPIO → LED / buzzer (driver if needed)	Shows state and reason
Power (bench)	Protected USB supply, common ground	Pump on separate appropriate supply; spill containment

Worked example (illustrative): zA = 2.00 m, dA = 0.80 m → hA = 1.20 m; zB = 2.10 m, dB = 1.00 m → hB = 1.10 m; Δh = 0.10 m. Raw dA − dB = −0.20 m would be wrong. Combined level uncertainty first estimate ≈ $\sqrt{(uA^2 + uB^2)}$; a Δh inside that is not meaningful.

Prior art we checked — and what each shows

Source	What it establishes	Limit of use
InnovateX 2026-27 brief (Techfest, IIT Bombay)	Theme 3 scope, Round 1 content, 10-min pitch, ~5,000 t/yr estimate	Organiser-quoted estimate, original survey not cited
Aquasition (2019) trash-screen monitoring	Single & differential level + camera monitoring with alarms exists	Vendor page; fixed screens, UK context
AMETEK app note: differential measurement, traveling screen	Two transmitters across screen → differential alarm / auto-clean	Industrial intake screens, not floating booms
IPT towing-basin trash-boom tests (IAHR library)	Line tension depends on flow speed and debris accumulation	Scale model; abstract-level access
Ocean Engineering (2022) trash-net mooring study	Mooring / net loads modelled for trash-blocking nets	Abstract only (full text not accessed)
Fluids (2022) trash-blocking net model	Net tension rises with width & solidity	Numerical + lab validation
Free Press Journal, 30 Jul 2026	BMC ₹14.5 cr, 2-yr O&M contract for booms on 6 rivers/nullahs	News report; no site list
JSN-SR04T & HX711 datasheets (robu.in)	Component interfaces and ranges	Supplier documents, not field durability

Test matrix and metric definitions

Factor	Levels	Count
Debris state	clean · partial · heavy (retrievable sample debris)	3
Flow setting	low · high (checked independently)	2
Downstream boundary	free · raised weir (backwater)	2
Repeats	physically reset, randomised order	5
Evaluation runs	held-out, rules frozen beforehand	60

Plus separate fault-injection and recovery runs (test E). Calibration runs are kept apart from evaluation runs.

Metric	Definition
False-positive rate	FP / (FP + TN)
Missed-event rate	FN / (FN + TP)
Detection delay	event onset → qualifying alert
Uncertain-output rate	UNCERTAIN outputs / all outputs
Valid-data fraction	valid samples / expected samples

Report raw counts and run conditions. Abstentions are not counted as correct. A 60-run bench test cannot support universal accuracy claims.

Budget assumptions and what is excluded

Bench planning item	INR
ESP32, two level modules, load cell, HX711, logging, wiring, sensor fixtures	3,400
Transparent channel + leak containment	1,800
Pump, hoses and reservoir	1,700
Model boom, mooring and frame	650
Adjustable weir / downstream level control	350
Reference rulers, weights and flow check	450
Fabrication and miscellaneous	300
Base total	**8,650**
15% reserve	**1,297.50**
Planned total (rounded from 9,947.50)	**9,950**

Not included (field costs, quote per site)

- Field-rated sensors and marine tension link
- Enclosure, corrosion protection, mounting
- Power (solar/battery) and communication
- Survey, installation, commissioning
- Cleaning, recalibration, replacement visits
- Crew time for handling alerts

Planning ceiling ₹12,000 → ~₹2,050 headroom.

Limitations and deployment risks

Risk	Consequence	Planned response
Foam, spray, turbulence under ultrasonic	Invalid or wrong range	Validity flags, mounting checks, challenge tests
Bracket / datum movement	False Δh	Survey reference checks, recalibrate after remount
Load-cell drift, fixture friction	Wrong tension	Zero/span, creep, hysteresis checks
Slack or shifted mooring	Misleading load	UNCERTAIN state + physical inspection
Tide, backwater, reverse flow	Ambiguous hydraulics	Site-specific operating envelope → UNCERTAIN
Shallow or angled boom	Δh too small to measure	Gate A: stop if inside uncertainty
Network / power / SD failure	Stale or missing data	Local buffer, stale flag, fault record
Corrosion, fouling (field)	Drift, lost access	Field-rated materials; service interval TBD
No crew available	Alert causes no action	Agree responsibility before any pilot